

PASSWORD SECURITY

STRONG PASSWORDS TODAY,
SECURE TOMORROW.

A strong password is your first line of defense against hackers. Good habits today protect your data, identity and digital life.



WHY PASSWORD SECURITY MATTERS?

- Weak passwords are easy to guess and crack.
- Compromised passwords lead to data breaches and financial loss.
- Reused passwords can put all your accounts at risk.
- Strong passwords keep your personal and professional life secure.

PASSWORD STRENGTH

Very Weak	123456	< 1 second
Weak	password	< 1 second
Medium	P@ssw0rd	1 minute
Strong	Tr0ub4dor&3	1 hour
Very Strong	ThisIs@VeryStr0ng2024	Years
Excellent	CorrectHorseBatteryStaple!	Centuries

HOW TO CREATE A STRONG PASSWORD



1. THINK UNIQUE

Don't use personal information or common words.

A a 1 !

2. USE A MIX

Use uppercase, lowercase, numbers and special characters.



3. MAKE IT LONG

Use at least 12-16 characters. Longer is always better.



4. AVOID PATTERNS

Don't use simple patterns like 123456 or qwerty or word1.



5. KEEP IT SECRET

Never share your password with anyone.



6. CHANGE WISELY

Change passwords regularly, especially for important accounts.

GOOD PASSWORD EXAMPLES

- ✓ Tr0ub4dor&3!
- ✓ C0ffee_Is#Life@2024
- ✓ B1ue\$ky*R0cks!
- ✓ L0ngP@ssw0rdIs\$ecure!
- ✓ N3v3r_Gu3ss_M3_2024!

BAD PASSWORD EXAMPLES

- ✗ 123456
- ✗ password
- ✗ qwerty
- ✗ admin123
- ✗ letmein
- ✗ welcome
- ✗ abc123
- ✗ password1
- ✗ 123456789

PASSWORD ANATOMY

ThisIs@Str0ngP@ss2024

Uppercase Letters

T, I, S, P

Lowercase Letters

h, s, s, t, r, o, n, g, s, s

Numbers

0, 2, 4

Special Characters

!, @, #

Length: 20 characters (Strong)

TIPS FOR PASSWORD SECURITY

- Use unique passwords for every account.
- Use a password manager to store passwords securely.
- Enable Two-Factor Authentication (2FA).
- Change passwords regularly.
- Beware of phishing emails and fake websites.
- Don't save passwords on public or shared devices.

COMMON PASSWORD ATTACKS

- Brute Force Attack**
Hackers try millions of password combinations until the correct one is found.
- Dictionary Attack**
Attackers use a list of common words and passwords to guess your password.
- Credential Stuffing**
Hackers use leaked usernames and passwords from other breaches to access your accounts.
- Phishing Attack**
Attackers trick you into giving your password through fake websites or emails.

PASSWORD MANAGER BENEFITS

- Generate strong, random passwords
- Store all passwords in encrypted vault
- Auto-fill passwords on websites and apps
- Access passwords from any device securely
- Reduces risk of password reuse



HOW PASSWORDS GET STOLEN



Phishing



Keyloggers



Data Breaches



Malware



Social Engineering



Unsecured Wi-Fi

2FA - ADD AN EXTRA LAYER

Enable Two-Factor Authentication wherever possible.



Password



Your Device



Access

Even if someone gets your password, they can't access without your device.

PASSWORD STRENGTH GUIDE

Length	Example	Strength	Time to Crack (Brute Force)
6	abc123	Very Weak	Instantly
8	password1	Weak	Seconds
10	P@ssw0rd10	Medium	Minutes
12	Tr0ub4dor&3!	Strong	Hours
16	C0mpl3x!P@ssw0rd2024	Very Strong	Years
20+	CorrectHorseBatteryStaple!	Excellent	Centuries

PASSWORD SECURITY CHECKLIST

- Use at least 12-16 characters
- Include uppercase, lowercase, numbers & symbols
- Avoid dictionary words and personal info
- Use unique passwords for all accounts
- Enable 2FA wherever available
- Use a password manager
- Update passwords regularly
- Keep your devices and software updated

QUICK REMINDERS

- Strong password = Strong protection
- Your password is like your toothbrush. Don't share it!
- A little effort today, saves you from big trouble tomorrow.



STRONG PASSWORDS.
STRONG SECURITY.



Protect your accounts.
Protect your future.

security@yourlife:~\$ secure --password
Access Granted. Stay Safe! 🛡️

PASSWORD CRACKING

EVERY PASSWORD HAS A WEAKNESS.
WE JUST FIND IT.

Password cracking is the process of recovering lost or forgotten passwords by using various techniques and tools to find the original password from its hashed form.



WHAT IS PASSWORD CRACKING?

Password cracking is the art of turning encrypted (hashed) passwords back into readable text using different methods and attacks.



COMMON HASH ALGORITHMS

- MD5
- SHA1
- SHA256 / SHA512
- bcrypt
- NTLM
- LM



HOW PASSWORD CRACKING WORKS



1. OBTAIN HASH

Get the password hash from system, website, database, or network.



2. CHOOSE METHOD

Select the best attack method for the hash type and scenario.



3. START ATTACK

Use tools to test millions or billions of password combinations.



4. PASSWORD FOUND

When the correct password matches the hash, it is revealed.



5. ACCESS GRANTED

Use the recovered password to gain authorized access or escalate privileges.

PASSWORD CRACKING METHODS



BRUTE FORCE ATTACK

Tries all possible combinations of characters until the correct password is found.

Slow but effective on simple passwords.



DICTIONARY ATTACK

Uses a list of common passwords and words to guess the password.

Fast and effective for weak passwords.



HYBRID ATTACK

Combines dictionary words with numbers, symbols, and variations.

Balance between speed and accuracy.



MASK ATTACK

Uses a pattern (mask) you define to reduce the search space.

Example: ?[?l]d?d
(2 letters + 2 digits)



RAINBOW TABLE ATTACK

Uses precomputed tables of hash-password pairs to quickly crack hashes.

Very fast, but needs large storage.



CREDENTIAL STUFFING

Uses leaked username:password pairs to try logging in to other websites.

Targets password reusing users.

POPULAR TOOLS



JOHN THE RIPPER

Powerful and flexible password cracking tool.



HASHCAT

World's fastest password recovery tool.



HYDRA

Network login cracker supporting many protocols.



MEDUSA

Brute force tool with parallel login cracking.



NCRACK

Part of Nmap project for network login brute forcing.

EXAMPLE: JOHN THE RIPPER

```
$ john --format=sha256 hash.txt --wordlist=rockyou.txt
```

Using default input encoding: UTF-8
Loaded 1 password hash (sha256)

Will run 4 OpenMP threads

Press 'q' or Ctrl-C to abort, almost any other key for status

```
letmein123 (?)
```

```
1g 0:00:00:00 DONE (2024-05-21 10:15) 100.0g/s 123456789..password
```

Use the "--show" option to display all of the cracked passwords reliably

```
$ john --show hash.txt
```

```
letmein123
```

```
1 password hash cracked, 1 password found
```

COMMON WORDLISTS

- rockyou.txt (Most popular)
- subdomains-top1million-110000.txt
- fasttrack.txt
- common.txt
- passwords.txt
- darkweb2017-top10000.txt



PASSWORD STRENGTH GUIDE

STRENGTH	EXAMPLE	TIME TO CRACK (BRUTE FORCE)
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Weak	password	< 1 second
Medium	P@ssw0rd	1 minute
Strong	Tr0ub4dor&3	1 hour
Very Strong	ThisIs@Very\$tr0ng2024	Years
Extreme	CorrectHorseBatteryStaple!	Centuries

HASH EXAMPLE

Password: letmein123

SHA256 Hash:

```
a4d90eac9ab26a4a2da04125bc2c096e  
f977f7e2b67f6b8c0d6b6d6b6dd6b66
```



HASH TYPES EXAMPLE

```
MD5      : 5f4dcc3b5aa765d61d8327deb882cf99  
SHA1     : e599f2a1ba31ecdd1ae84f75c0aa474f3a663f05f4  
SHA256   : a4d90eac9ab26a4a2da04125bc2c096ef977f7e2...  
NTLM     : aad3b435b51404eeaad3b435b51404ee  
bcrypt   : $2y$10$KYYbZ5JFVfq0v9BlnF5u0C3fG8e1eT...
```



TIPS FOR DEFENDERS

- ☑ Use strong, unique passwords
- ☑ Enable Two-Factor Authentication (2FA)
- ☑ Use slow hashing algorithms (bcrypt, Argon2)
- ☑ Add salt to passwords
- ☑ Monitor login attempts and log suspicious activity
- ☑ Educate users about password security

COMMON USE CASES



Penetration
Testing



Security
Auditing



Forensic
Analysis



Password
Recovery



DISCLAIMER

This material is for educational and authorized penetration testing purposes only. Unauthorized access to systems is illegal and punishable by law.



“ Strong passwords keep hackers out.
Stronger knowledge keeps you ahead.



KNOWLEDGE IS POWER.
USE IT ETHICALLY.

```
root@kali:~#
```

```
root@kali:~# █
```

BRUTE FORCE ATTACK

THE POWER OF TRIAL AND ERROR.
ONE ATTEMPT CLOSER TO **BREAKING IN**.

A Brute Force Attack is a method used by attackers to gain unauthorized access by trying a large number of possible passwords or keys until the correct one is found.



LOGIN

Username
admin

Password

Trying: password123
Status: **FAILED**

Trying: qwerty2024
Status: **FAILED**

Trying: admin@123
Status: **SUCCESS**



GOAL:
Find the correct password and gain access.

HOW BRUTE FORCE ATTACK WORKS



1. TARGET

Attacker selects a target system or service.



2. GATHER INFO

Collect information like usernames, email, etc.



3. CREATE LIST

Create or load a list of possible passwords.



4. ATTEMPT LOGIN

Try each password automatically against the target.



5. CHECK RESPONSE

Check the server response for success or failure.



6. ACCESS GRANTED

Correct password found. Access granted!

WHERE CAN IT BE USED?

- SSH Login (22)
- FTP Login (21)
- Web Login Forms (HTTP/HTTPS)
- RDP Login (3389)
- Email Login (IMAP/POP3)
- Wi-Fi Password (WPA/WPS)
- Database Login (MySQL, PostgreSQL)

TYPES OF BRUTE FORCE ATTACKS



SIMPLE BRUTE FORCE

Tries every possible combination one by one.



DICTIONARY ATTACK

Uses a list of common passwords.



HYBRID ATTACK

Combines dictionary words with variations (e.g., numbers, symbols).



REVERSE BRUTE FORCE

Tries one password against many usernames.



CREDENTIAL STUFFING

Uses leaked username:password pairs to gain access.

POPULAR TOOLS

TOOL	DESCRIPTION
 Hydra	Fast, reliable and very flexible login cracker
 Medusa	Parallel login brute forcer with many modules
 Ncrack	Part of Nmap project. Network login cracker
 Patator	Multi-purpose brute forcer (very powerful)
 Burp Suite	Used for brute forcing web login forms
 John the Ripper	Password cracker and security auditing tool

EXAMPLE: HYDRA SSH BRUTE FORCE

Command

```
hydra -l admin -P passwords.txt ssh://192.168.1.10
```

Options:

```
-l <user> : Username  
-P <file> : Password list file  
ssh://<ip> : Target IP and service
```

Output (Example):

```
Hydra v9.4 (c) 2024 by van Housen/THC & David Maciejak - Please do not  
use in military or secret service organizations, or for illegal purposes.  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2024-05-21 12:30:10  
[DATA] max 16 tasks per 1 server, max 16 tasks, 160 login tries (1/1/160), 7 tries per task  
[DATA] attacking ssh://192.168.1.10:22/  
[22][ssh] host: 192.168.1.10 login: admin password: admin@123  
1 of 1 target successfully completed, 1 valid password found  
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2024-05-21 12:30:10
```

REAL WORLD EXAMPLE (WEB LOGIN BRUTE FORCE)



HOW TO PREVENT BRUTE FORCE ATTACKS

-  Use Strong Passwords
-  Implement Account Lockout
-  Use IP Whitelisting
-  Enable Two-Factor Authentication
-  Limit Login Attempts
-  Monitor Logs and Alerts
-  Use Firewall and IDS/IPS

BEST PRACTICES

- ✓ Use complex passwords (letters, numbers, symbols)
- ✓ Change default usernames and passwords
- ✓ Keep systems and software up to date
- ✓ Use CAPTCHA on login forms
- ✓ Enable multi-factor authentication (MFA)
- ✓ Regularly review and monitor login logs



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KNOWLEDGE IS POWER.
USE IT ETHICALLY.

- ✓ Learn
- ✓ Practice in legal labs
- ✓ Protect
- ✓ Make the internet safer



WEBSITE SECURITY

 **SECURE TODAY, SAFE TOMORROW**

Website security protects your site, your users and your data from cyber threats and attacks.

WHY WEBSITE SECURITY MATTERS

-  Protects user data and privacy
-  Builds trust and improves reputation
-  Prevents data breaches and financial loss
-  Improves SEO rankings (Google prefers secure sites)
-  Ensures website availability and uptime

COMMON WEBSITE THREATS

SQL INJECTION



Attackers inject malicious SQL code to access or manipulate your database.

CROSS-SITE SCRIPTING (XSS)



Malicious scripts are injected into trusted websites and executed in users' browsers.

CROSS-SITE REQUEST FORGERY (CSRF)



Tricks users into performing unwanted actions on a web application where they are authenticated.

BROKEN AUTHENTICATION & SESSION



Weak login, session hijacking or poor credential management can lead to unauthorized access.

SECURITY MISCONFIGURATION



Improper server, framework or plugin configurations can expose vulnerabilities.

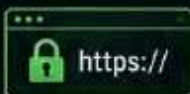
FILE INCLUSION VULNERABILITIES



Attackers include malicious files to execute code on your server.

BEST PRACTICES TO SECURE YOUR WEBSITE

USE HTTPS



Always use SSL/TLS certificates to encrypt data between users and your website.

KEEP EVERYTHING UPDATED



Keep your CMS, plugins, themes, and server software up to date.

USE STRONG AUTHENTICATION



Enforce strong passwords, two-factor authentication (2FA) and limit login attempts.

VALIDATE & SANITIZE INPUTS



Validate and sanitize all user inputs to prevent SQL Injection, XSS and other attacks.

BACKUP YOUR WEBSITE



Take regular backups and store them securely offsite.

USE WEB APPLICATION FIREWALL (WAF)



WAF helps block malicious traffic and common attack patterns.

SECURITY HEADERS YOU SHOULD ENABLE



Content-Security-Policy	Prevents XSS and data injection attacks
X-Frame-Options	Protects against clickjacking attacks
X-Content-Type-Options	Prevents MIME type sniffing
Strict-Transport-Security	Enforces HTTPS connections
Referrer-Policy	Controls referrer information
Permissions-Policy	Restricts access to browser features

SECURE DEVELOPMENT TIPS

- ✓ Follow OWASP Top 10 guidelines
- ✓ Use secure coding practices
- ✓ Implement least privilege access
- ✓ Disable directory listing
- ✓ Hide error messages from users
- ✓ Test for vulnerabilities regularly

OWASP TOP 10 (2021)

- 1 Broken Access Control
- 2 Cryptographic Failures
- 3 Injection
- 4 Insecure Design
- 5 Security Misconfiguration
- 6 Vulnerable and Outdated Components
- 7 Identification and Authentication Failures
- 8 Software and Data Integrity Failures
- 9 Security Logging and Monitoring Failures
- 10 Server-Side Request Forgery (SSRF)

HOW ATTACKERS HACK A WEBSITE (TYPICAL FLOW)



RECONNAISSANCE
Gather information about the target website.



SCAN & ENUMERATE
Scan for open ports, subdomains and vulnerabilities.



FIND VULNERABILITIES
Identify security flaws in code, plugins or configurations.



EXPLOIT
Exploit the vulnerability to gain access.



COMPROMISE
Steal, modify or destroy data or deface the website.

USEFUL SECURITY TOOLS

 OWASP ZAP	Web application security scanner
 Burp Suite	Web penetration testing tool
 Nmap	Network scanning and enumeration
 Nikto	Web server scanner
 SQLMap	Automates SQL injection attacks
 WPScan	WordPress security scanner



SECURITY IS NOT A FEATURE, IT'S A FOUNDATION.

**PROTECT YOUR WEBSITE.
PROTECT YOUR USERS.
PROTECT YOUR FUTURE.**



JOHN THE RAPPERS

THE LEGENDARY HACKER WHO RAPED THE SYSTEM

John the Rippers is a fast password cracker, currently available for many platforms, designed to detect weak Unix passwords.



"I'm not a criminal,
I'm a security
researcher.
I break to learn,
not for profit."
- John the Rappers

WHO IS JOHN THE RAPPERS?

John the Rappers is a fictional character and the mascot of the John the Ripper password cracking software.



The name is a play on words of "Jack the Ripper", the infamous 19th-century killer.

KEY FEATURES

- Multi-platform support (Windows, Linux, macOS, etc.)
- Supports many hash types
- High performance
- Customizable & modular
- Active development for over 2 decades

HOW JOHN THE RIPPERS WORKS



1. GET HASH

Obtain password hash from target system.



2. CHOOSE MODE

Select the appropriate attack mode.



3. START CRACKING

Try millions/billions of passwords per second.



4. PASSWORD FOUND

When a match is found, password is revealed.



5. ACCESS GRANTED

Use the password to gain authorized access.

ATTACK MODES

Dictionary Attack	Uses a wordlist to guess passwords
Brute Force Attack	Tries all possible combinations
Incremental Attack	Starts from a set and increments
Hybrid Attack	Combines dictionary + rules
Mask Attack	Cracks based on a mask pattern

SUPPORTED HASH TYPES (Examples)

✓ MD5	✓ DES
✓ SHA-1	✓ BSDI
✓ SHA-256	✓ Kerberos
✓ bcrypt	✓ Oracle
✓ NTLM	✓ ZIP
✓ LM	✓ And many more...

```
john@kali:~$ john hash.txt --wordlist=rockyou.txt
Using default input encoding: UTF-8
Loaded 1 password hash (SSHA [5SHA512 128/128 AVX 2x])
Cost 1 (iteration count) is 5000 for all loaded hashes
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
letmein123 (7)
lg 0:00:00:00 DONE (2024-05-21 18:15) 100.0g/s 26480p/s 38480c/s
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
john@kali:~$ john --show hash.txt
user: letmein123
```

COMMON USE CASES



Password
Auditing



Penetration
Testing



Security
Research



Education &
Training



CTF & Capture
the Flag

TIPS FOR ETHICAL USE



Get proper
authorization



Use only in legal
and permitted
environments



Respect privacy
and data
protection



Report vulnerabilities
responsibly



Use for learning,
improving security
and helping others



DISCLAIMER

This tool is for educational and ethical purposes only. Unauthorized access to systems is illegal and unethical.



WITH GREAT POWER
COMES GREAT RESPONSIBILITY.



ETHICAL HACKING
BUILDS A SECURE WORLD.



STAY LEGAL.
STAY ETHICAL.
HACKING IS NOT A CRIME,
BUT UNAUTHORIZED
HACKING IS.

SQLMAP

Automatic SQL Injection and Database Takeover Tool

sqlmap is an open-source penetration testing tool that automates the process of **detecting and exploiting SQL injection flaws** and taking over database servers.



KEY FEATURES

- Automatic detection of SQL injection flaws
- Support for multiple databases
- Full database takeover capability
- Data fetching from the database
- Built-in database management
- OS command execution
- User-friendly and powerful

HOW SQLMAP WORKS



COMMON SQLMAP COMMANDS

COMMAND	DESCRIPTION	EXAMPLE
Detect SQLi	Detects SQL injection vulnerabilities	<code>sqlmap -u "http://target.com/page.php?id=1"</code>
List Databases	Enumerate all databases	<code>sqlmap -u "URL" --dbs</code>
List Tables	Enumerate tables in a database	<code>sqlmap -u "URL" -D database_name --tables</code>
List Columns	Enumerate columns in a table	<code>sqlmap -u "URL" -D db -T table --columns</code>
Dump Data	Dump data from a table	<code>sqlmap -u "URL" -D db -T table --dump</code>
Get Current DB	Get the current database	<code>sqlmap -u "URL" --current-db</code>
OS Shell	Get OS command shell	<code>sqlmap -u "URL" --os-shell</code>
File Read	Read files from the server	<code>sqlmap -u "URL" --file-read /etc/passwd</code>
File Write	Write files to the server	<code>sqlmap -u "URL" --file-write shell.php --file-dest /var/www/html/</code>
Scan All	Full scan with maximum tests	<code>sqlmap -u "URL" --level 5 --risk 3 --batch</code>
Update/Check	Update sqlmap / Check version	<code>sqlmap --update</code> <code>sqlmap --version</code>

BASIC USAGE

```
sqlmap -u "http://target.com/page.php?id=1" --dbs
```

- `-u` Target URL
- `--dbs` Enumerate all databases
- `-D` Use a specific database
- `-T` Use a specific table
- `--dump` Dump data from table
- `--batch` Never ask for user input
- `--level` Set testing level (1-5)
- `--risk` Set risk level (1-3)
- `--threads` Set number of threads



DATABASES SUPPORTED



And many more...

SQLMAP WORKFLOW (TYPICAL)



EXAMPLE WORKFLOW (DEMO)

```
$ sqlmap -u "http://testphp.vulnweb.com/listproducts.php?cat=1" --dbs
[*] testing connection to the target URL
[*] sql injection vulnerability detected
available databases (5):
[*] acuart
[*] information_schema
[*] mysql
[*] testphp
[*] vulnweb
[*] fetched data logged to text files under '/home/user/.sqlmap/output/'
[*] shutting down at 18:35:47
```

TIPS & BEST PRACTICES

- Always get permission before testing.
- Use `--batch` to automate non-interactive mode.
- Use `--level` and `--risk` carefully for deep testing.
- Check logs in `~/.sqlmap/output/` for results.
- Keep sqlmap updated for the latest features.
- Use a VPN and test in authorized environments.

LEGAL DISCLAIMER

sqlmap is provided for educational and authorized penetration testing only. Unauthorized use is illegal. You are responsible for your actions.

QUICK CHEAT SHEET

Enumerate DBs	<code>sqlmap -u "URL" --dbs</code>
Enumerate Tables	<code>sqlmap -u "URL" -D db --tables</code>
Enumerate Columns	<code>sqlmap -u "URL" -D db -T tbl --columns</code>
Dump Data	<code>sqlmap -u "URL" -D db -T tbl --dump</code>
OS Shell	<code>sqlmap -u "URL" --os-shell</code>
Read File	<code>sqlmap -u "URL" --file-read /path/to/file</code>
Write File	<code>sqlmap -u "URL" --file-write shell.php --file-dest /var/www/html/</code>
Full Scan	<code>sqlmap -u "URL" --level 5 --risk 3 --batch</code>

>_ KNOW THE DATABASE. OWN THE SYSTEM.

LEARN ETHICALLY. HACK RESPONSIBLY.



NETWORK SCAN

DISCOVER. ENUMERATE. EXPLOIT.

Network scanning is the first step in any penetration test. It helps identify live hosts, open ports, running services, and potential vulnerabilities.



OBJECTIVES OF NETWORK SCANNING

- Discover live hosts on the network
- Identify open ports and services
- Detect operating systems
- Find vulnerabilities and misconfigurations
- Map the network and its topology
- Gather information for further attacks

NETWORK SCANNING WORKFLOW



POPULAR TOOLS

	Nmap	Network scanner and security auditor
	Masscan	Mass IP port scanner, extremely fast
	Netdiscover	ARP based network scanner
	Angry IP Scanner	Fast IP and port scanner with GUI
	Zenmap	Nmap GUI for easier scanning and analysis

TYPES OF SCANS (NMAP)

TCP SYN Scan (-sS)	Stealthy scan, half-open scan. Default scan in Nmap.	
TCP Connect Scan (-sT)	Full TCP connection scan. More reliable but less stealthy.	
UDP Scan (-sU)	Find open UDP ports. Slower but necessary.	
Ping Sweep (-sn)	Discover live hosts without port scanning.	
OS Detection (-O)	Detect operating system of target host.	
Version Detection (-sV)	Detect service versions on open ports.	

BASIC NMAP SCAN EXAMPLES

```
# Ping sweep - Find live hosts
nmap -sn 192.168.1.0/24

# TCP SYN scan
nmap -sS 192.168.1.10

# TCP Connect scan
nmap -sT 192.168.1.10

# UDP scan
nmap -sU 192.168.1.10

# Scan all ports
nmap -p- 192.168.1.10

# Scan top 1000 ports
nmap --top-ports 1000 192.168.1.10

# Version detection
nmap -sV 192.168.1.10

# OS detection
nmap -O 192.168.1.10

# Aggressive scan (OS, version, script)
nmap -A 192.168.1.10

# Save results to file
nmap -A 192.168.1.10 -oN scan.txt
```

USEFUL NMAP FLAGS

-sn	Ping scan (host discovery)
-sS	TCP SYN scan (stealth scan)
-sT	TCP Connect scan
-sU	UDP scan
-sV	Version detection
-O	OS detection
-p-	Scan all 65535 ports
-p <ports>	Scan specific ports (e.g., -p 22,80,443)
-A	Aggressive scan (OS, version, script, traceroute)
-T0 to -T5	Timing templates (0=slow, 5=insane)
-oN	Normal output to file
-oX	XML output to file
--script <script>	Run NSE script

NMAP TIMING TEMPLATES

-T0	Paranoid	Very slow, IDS evasion
-T1	Sneaky	Slow, IDS evasion
-T2	Polite	Slower, less intrusive
-T3	Normal	Default timing
-T4	Aggressive	Faster scan
-T5	Insane	Very fast, may crash targets

EXAMPLE OUTPUT (NMAP)

```
$ nmap -sS -sV -O 192.168.1.10
Starting Nmap 7.94 ( https://nmap.org ) at 2024-05-21 12:36
Nmap scan report for 192.168.1.10
Host is up (0.00045s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
22/tcp    open  ssh          OpenSSH 8.9p1 Ubuntu 8ubuntu0.6 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http         Apache/2.4.52 ((Ubuntu))
443/tcp   open  ssl/http     Apache/2.4.52 ((Ubuntu))
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel
Device type: general purpose
Running: Linux 5.X|6.X
OS CPE: cpe:/o:linux:linux_kernel:5 cpe:/o:linux:linux_kernel:6
OS details: Linux 5.15 - 6.6
Network Distance: 1 hop
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 12.45 seconds
```

PORT STATES

	open	Port is open and accepting connections
	closed	Port is closed
	filtered	Port is blocked by firewall or filter
	unfiltered	Port is open or filtered, Nmap can't decide
	open filtered	Could be open or filtered
	closed filtered	Could be closed or filtered

COMMON PORTS

PORT	SERVICE	DESCRIPTION
20	FTP-DATA	File Transfer [Data]
21	FTP	File Transfer Protocol
22	SSH	Secure Shell
23	TELNET	Telnet Remote Login
25	SMTP	Simple Mail Transfer
53	DNS	Domain Name System
80	HTTP	HyperText Transfer
110	POP3	Post Office Protocol v3
143	IMAP	Internet Message Access
443	HTTPS	HTTP Secure (SSL/TLS)
3306	MYSQL	MySQL Database
3389	RDP	Remote Desktop Protocol

BEST PRACTICES

- Always get authorization before scanning networks.
- Use appropriate timing (-T) to avoid detection.
- Combine scans with enumeration and vulnerability assessment.
- Save and analyze results carefully.
- Don't scan networks or systems you don't own.
- Keep tools and databases updated.

DISCLAIMER

This material is for educational purposes only. Unauthorized scanning is illegal. Use responsibly and ethically.



WEBPAGE ATTACK

Attack the Web. Know the Threats. Secure the Future.



WHAT IS WEBPAGE ATTACK?

A webpage attack targets vulnerabilities in a website or web application to steal data, deface pages, disrupt services, or take unauthorized control of the system.

ATTACKER GOALS



Steal Data



Bypass Authentication



Deface Website



Gain Unauthorized Access



Disrupt Services (DoS)

COMMON WEBPAGE ATTACKS

1. SQL INJECTION (SQLi)



Injecting malicious SQL queries to access, modify or delete database data.

Example: `' OR '1'='1;--`

2. CROSS-SITE SCRIPTING (XSS)



Injecting malicious scripts into web pages viewed by other users.

Example: `<script>alert(1)</script>`

3. CROSS-SITE REQUEST FORGERY (CSRF)



Forcing users to execute unwanted actions on a web application where they are authenticated.

4. BROKEN AUTHENTICATION



Exploiting flaws in login mechanisms to gain unauthorized access.

Example: Brute force login

5. FILE INCLUSION (LFI / RFI)



Including local or remote files to execute malicious code on the server.

Example: `../../../../etc/passwd`

6. COMMAND INJECTION



Executing system commands on the server through vulnerable input fields.

Example: `ls -la`

HOW A WEBPAGE ATTACK WORKS (TYPICAL FLOW)



1. RECONNAISSANCE

Attacker gathers info about the website, technologies, and vulnerabilities.



2. SCANNING

Scan for open ports, directories, parameters and known vulnerabilities.



3. FIND VULNERABILITY

Identify weak points in code, configuration or components.



4. EXPLOITATION

Exploit the vulnerability to execute malicious payload.



5. GAIN ACCESS

Gain unauthorized access to data, files or server.



6. MAINTAIN ACCESS

Install backdoors and maintain persistence for future access.



7. ACTIONS ON OBJECTIVES

Steal data, deface website, disrupt services or escalate privileges.

EXAMPLE: SQL INJECTION ATTACK

Vulnerable Login Page

Username:

admin

Password:

`' OR '1'='1;--`

Login

Result

✓ Login Successful!
Access Granted.



Database Compromised

EXAMPLE: XSS ATTACK

Vulnerable Comment Box

Write a comment...

Submit

Malicious Script Executes

```
<script>
alert("XSS Attack!");
</script>
```



TOOLS USED BY ATTACKERS

	Burp Suite	Web application security testing
	OWASP ZAP	Web vulnerability scanner
	SQLMap	Automates SQL injection attacks
	Nmap	Network scanning tool
	Nikto	Web server vulnerability scanner
	Dirb / Gobuster	Directory & file brute forcing
	Metasploit	Exploitation framework

HOW TO PROTECT YOUR WEBPAGE



Validate & sanitize all user inputs.



Use prepared statements to prevent SQL injection.



Implement strong authentication & authorization.



Use HTTPS and secure cookies.



Keep software, plugins & frameworks updated.



Use Web Application Firewall (WAF).



Monitor logs & detect suspicious activities.



Regularly backup your website and database.



WARNING

Unauthorized hacking is illegal. Use this knowledge for educational and defensive purposes only.



SECURE YOUR CODE.
SECURE YOUR WORLD.



- Secure Today
- Protect Tomorrow
- Stay One Step Ahead

WPS ATTACK

(Wi-Fi Protected Setup Attack)



WPS is a network security standard that aims to make it easy to set up a secure Wi-Fi network. But a design flaw can be exploited by attackers to gain access.



WARNING

This information is provided for educational purposes only. Use responsibly and only on networks you own or have permission to test.

WHAT IS WPS?

WPS helps users connect devices to a Wi-Fi network without entering the password manually. It uses an 8-digit PIN or Push Button method.

The PIN method has a vulnerability that makes brute-force attacks possible.



WPS MODES

1234
5678

PIN Method

Uses an 8-digit PIN to authenticate the client. This is vulnerable to brute-force attacks.



Push Button Configuration (PBC)

User presses a physical button on the router and client to connect. Less vulnerable but can have flaws in implementation.



HOW WPS PIN ATTACK WORKS



- 1 Attacker enables monitor mode on wireless adapter and scans for target networks.



- 2 Attacker gets the target's WPS PIN (usually the first half is known or brute-forced).



- 3 Attacker sends PIN guesses to the router and checks responses.



- 4 If the PIN is correct, the router accepts the request.



- 5 Attacker gains access to the Wi-Fi network and can intercept or manipulate data.

WHY WPS PIN IS VULNERABLE

- 8-digit PIN = 10,000,000 possible combinations
- Router reveals whether the first half is correct or not
- Attackers can brute-force the PIN offline or online
- Most routers do not lock out after many failed attempts



TOOLS USED

```
>_
```

- Reaver
- Bully (for Pixie Dust attack)
- Aircrack-ng suite
- Wash (for scanning WPS networks)
- Airmon-ng
- Ralink/Realtek compatible adapters

Note: Some routers are not vulnerable.

EXAMPLE COMMAND (REAVR)

```
reaver -i wlan0mon -b <BSSID> -vv
```

Options:

- i Monitor mode interface
- b Target BSSID (MAC address of router)
- vv Verbose output

Example:

```
reaver -i wlan0mon -b 00:11:22:33:44:55 -vv
```

WPS ATTACK METHODS

METHOD	DESCRIPTION	SPEED	REQUIREMENTS
BRUTE FORCE (PIN)	Tries all possible PIN combinations until the correct one is found.	Slow to Medium	WPS enabled router, monitor mode adapter
PIXIE DUST (Offline)	Captures handshake and cracks PIN offline. Much faster than online brute force.	Fast	Vulnerable router, supports Pixie Dust attack
PUSH BUTTON ATTACK	Exploits flaws in PBC implementation to get authentication.	Medium	Router with vulnerable PBC implementation

PROTECT YOUR NETWORK

- ✓ Disable WPS in your router settings (Recommended).
- ✓ Use strong WPA2/WPA3 password.
- ✓ Keep router firmware updated.
- ✓ Enable MAC filtering (optional).
- ✓ Monitor connected devices regularly.



WPS STATUS CHECK

You can check if WPS is enabled on your router. If enabled, disable it immediately.



KNOWLEDGE IS POWER. USE IT ETHICALLY.
SECURE YOUR NETWORK. PROTECT YOUR DATA.



WiFi ATTACK

Understanding. Protecting. Staying Secure.

 **UNAUTHORIZED ACCESS**
CAN STEAL YOUR DATA & PRIVACY

WHAT IS WiFi ATTACK?

WiFi attacks are attempts by attackers to gain unauthorized access to wireless networks, steal data, or disrupt services.



COMMON GOALS OF ATTACKERS

-  Steal passwords & sensitive data
-  Monitor network traffic
-  Access private networks
-  Inject malicious content

COMMON WiFi ATTACKS

1. EVIL TWIN ATTACK



Attacker creates a fake WiFi network with the same name to trick users. All traffic can be monitored.

2. PACKET SNIFFING



Captures data packets transmitted over the network and steals sensitive information.

3. DEAUTHENTICATION ATTACK



Sends deauth packets to disconnect users from the network and performs other attacks.

4. WPS PIN ATTACK



Exploits the WPS feature to brute-force the PIN and gain access to the WiFi network.

5. KRACK ATTACK



Exploits vulnerabilities in WiFi (WPA2) protocol to intercept and decrypt data.

HOW WiFi ATTACK WORKS (EXAMPLE: EVIL TWIN)



1. Attacker scans for nearby WiFi networks.



2. Attacker sets up a fake WiFi with the same name (Evil Twin).



3. Victim connects to the fake WiFi, thinking it's the real one.



4. Attacker intercepts traffic, steals data or credentials.



5. Data is misused for malicious activities.

TOOLS USED BY ATTACKERS

- Kali Linux
- Aircrack-ng
- Wireshark
- Reaver (WPS Attack)
- Airodump-ng
- Ettercap
- Bettercap



INFORMATION AT RISK

-  Usernames & Passwords
-  Emails & Messages
-  Browsing History
-  Cookies & Tokens
-  Credit Card Information

HOW TO PROTECT YOUR WiFi

- ✓ Use strong, unique passwords (WPA3 if possible)
- ✓ Disable WPS
- ✓ Keep router firmware updated
- ✓ Use a secure encryption (WPA2/WPA3)
- ✓ Enable MAC filtering
- ✓ Avoid connecting to unknown WiFi networks
- ✓ Use VPN for extra security



ATTACKER'S PROCESS (KILL CHAIN)



RECONNAISSANCE
Find target networks



SCANNING
Identify vulnerabilities



GAIN ACCESS
Exploit & connect



MAINTAIN ACCESS
Stay connected



EXPLOITATION
Steal or manipulate data



COVER TRACKS
Hide activities



WiFi is convenient,
but **not always safe.**

BE AWARE. BE SECURE. STAY PROTECTED.
Your security is in your hands.



MAN IN THE MIDDLE ATTACK



When the attacker sits in between,
trust is broken, data is exposed.



VICTIM / USER

ATTACKER

TARGET / SERVER



HOW MAN IN THE MIDDLE ATTACK WORKS



1. RECONNAISSANCE

Attacker identifies the target and the network.



2. INTERCEPTION

Attacker positions himself between the user and the server.



3. RELAY

Attacker relays messages between both sides to maintain the connection.



4. EAVESDROPPING

Attacker reads sensitive data; steals information or alters the data.



5. COVER TRACKS

Attacker may hide activities to avoid detection.

COMMON TECHNIQUES

- ARP Spoofing
- DNS Spoofing
- Wi-Fi Eavesdropping (Fake AP)
- SSL Stripping
- Session Hijacking



WHAT CAN BE STOLEN?

- 🔒 Usernames & Passwords
- 💳 Credit Card Information
- ✉️ Emails & Messages
- 🍪 Cookies & Session Tokens
- 📄 Sensitive Business Data

HOW TO DETECT

- Unexpected SSL certificate warnings
- Slow network or unusual delays
- Unusual ARP or DNS activity
- Tools: Wireshark, Netstat, ARP Table
- Intrusion Detection Systems (IDS)

EXAMPLE SCENARIO



HOW TO PREVENT

- 🛡️ Always use HTTPS
- 🛡️ Use VPN on public networks
- 🛡️ Enable Two-Factor Authentication (2FA)
- 🛡️ Verify website certificates
- 🛡️ Keep your system and software updated
- 🛡️ Avoid unsecured Wi-Fi networks



PROTECT YOURSELF. PROTECT YOUR DATA.

Awareness is your best defense against
Man in the Middle Attacks.



THINK BEFORE YOU CONNECT.
STAY ALERT. STAY SECURE.

PENETRATION TESTING

FIND IT. EXPLOIT IT. FIX IT. SECURE IT.

Penetration Testing is an authorized simulated cyber attack on a system, application or network to identify vulnerabilities that attackers could exploit.



GOALS

- ✓ Identify vulnerabilities
- ✓ Assess security posture
- ✓ Test effectiveness of controls
- ✓ Prevent real-world attacks
- ✓ Provide remediation guidance

THE PENETRATION TESTING PROCESS



COMMON ATTACK VECTOR



PENETRATION TESTING TYPES



TOOLS OF THE TRADE



RISK SEVERITY LEVELS



KILL CHAIN VIEW (EXAMPLE)



SECURITY IS NOT A PRODUCT, IT'S A PROCESS.



TEST TODAY, PROTECT TOMORROW.



WiFi Scan

Scan. Analyze. Secure.



STAY SECURE
KNOW YOUR NETWORK

NETWORKS FOUND

12

ENCRYPTED

8

OPEN NETWORKS

4

SCAN TIME

00:00:07

AVAILABLE NETWORKS

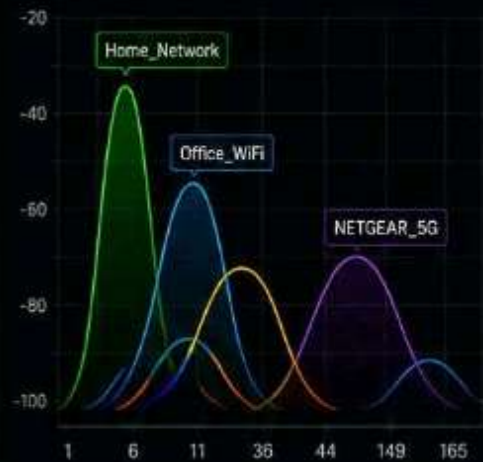
SCANNING...

SSID	CHANNEL	SIGNAL	SECURITY
Home_Network	6	-32 dBm	WPA2
Office_WiFi	11	-45 dBm	WPA2
CoffeeShop_Free	1	-67 dBm	OPEN
TP-Link_4G3F	6	-72 dBm	WPA/WPA2
AndroidAP_1234	11	-75 dBm	WPA2
Guest_Network	3	-80 dBm	OPEN
NETGEAR_5G	36	-55 dBm	WPA2
Hidden_Network	—	-82 dBm	WPA2
My_WiFi_Ext	9	-68 dBm	WPA/WPA2
Public_WiFi	1	-81 dBm	OPEN
D-Link_DIR-615	6	-71 dBm	WPA2
FiberNet_2024	44	-60 dBm	WPA2



SCAN NETWORKS

SIGNAL STRENGTH



CHANNEL OVERVIEW

2.4 GHz

5 GHz



BEST CHANNEL

2.4 GHz

6

5 GHz

36

NETWORK DETAILS

	SSID	Home_Network
	BSSID	1A:2B:3C:4D:5E:6F
	MAC VENDOR	TP-Link Technologies Co., Ltd.
	CHANNEL	6 (2.437 GHz)
	SECURITY	WPA2-PSK (AES)
	SIGNAL	-32 dBm (Excellent)
	LAST SEEN	Now



SECURITY CHECK

WI-FI SECURITY: STRONG.

This network is using WPA2 encryption. Keep your router firmware up to date.



TIPS

- ✓ Use WPA2 or WPA3 for better security.
- ✓ Change default router credentials.
- ✓ Hide your SSID to avoid unwanted access.
- ✓ Use strong passwords.
- ✓ Keep your router firmware updated.



WiFi Scan



Tools



Channels



History



Settings

CYBER SECURITY EXPLOIT

Turning Vulnerabilities Into Access

WHAT IS AN EXPLOIT?

An exploit is a piece of code or technique that takes advantage of a vulnerability in a system, application, or network to gain unauthorized access or cause damage.



EXPLOITS DON'T HAPPEN
BY ACCIDENT,
THEY HAPPEN BY
UNCOVERING WEAKNESS.

VULNERABILITY

CVE-2024-XXXX
Severity: HIGH

EXPLOIT ACTIVATED

Injecting payload...
75%

ACCESS GRANTED

Privileges: Administrator
System Control: Enabled



SYSTEM EXPLOITED

EXPLOIT LIFECYCLE



1. RECONNAISSANCE

Gather information about the target to find potential weaknesses.



2. SCANNING

Scan systems and services to identify vulnerabilities.



3. EXPLOITATION

Use an exploit to take advantage of the vulnerability.



4. ACCESS GAINED

Gain unauthorized access to the system.



5. MAINTAIN ACCESS

Maintain access, escalate privileges, or move laterally within the network.

COMMON EXPLOITS

- SQL Injection
- Cross-Site Scripting (XSS)
- Remote Code Execution (RCE)
- Buffer Overflow
- Privilege Escalation
- Zero-Day Exploits



PREVENTION & DEFENSE

- Keep systems and software updated
- Use firewalls and intrusion detection
- Follow the principle of least privilege
- Secure code and input validation
- Regular security audits & penetration testing
- Employee security awareness



SECURITY IS NOT A PRODUCT, IT'S A PROCESS.
THINK LIKE AN ATTACKER, DEFEND LIKE A PROTECTOR.

ETHICAL HACKING

HACKING WITH A PURPOSE, SECURING THE FUTURE



RECONNAISSANCE

- Footprinting
- Scanning
- Enumeration



MAINTAIN ACCESS

- Persistence
- Privilege Escalation
- Backdoors



GAIN ACCESS

- Exploitation
- Password Attacks
- Social Engineering



COVER TRACKS

- Clearing Logs
- Hiding Tracks
- Anonymity



THINK
LIKE A HACKER



FIND
VULNERABILITIES



PROTECT
AND SECURE



BUILD A
SAFER WORLD

ETHICAL HACKING IS NOT A CRIME
IT'S A RESPONSIBILITY

CYBER SECURITY

BE AWARE. BE SECURE. BE SAFE.

COMMON THREATS



PHISHING

Fraudulent attempts to steal sensitive information.



MALWARE

Malicious software designed to harm or exploit systems.



RANSOMWARE

Malware that locks your data and demands payment.



PASSWORD ATTACKS

Brute force or dictionary attacks to guess your passwords.



SOCIAL ENGINEERING

Manipulating people to disclose confidential information.

BEST PRACTICES



USE STRONG PASSWORDS

Use long, unique passwords and a password manager.



ENABLE 2FA

Add an extra layer of security to your accounts.



KEEP SOFTWARE UPDATED

Regular updates fix vulnerabilities and improve security.



THINK BEFORE YOU CLICK

Be cautious of links, attachments and unknown senders.



BACKUP YOUR DATA

Regular backups can save you from data loss or attacks.

THE CIA TRIAD



CONFIDENTIALITY

Ensure information is accessible only to authorized users.



INTEGRITY

Protect data from unauthorized changes or tampering.



AVAILABILITY

Ensure systems and data are available when needed.



REMEMBER

Cyber security starts with you. Stay informed, stay alert, stay secure.

“

“CYBER SECURITY IS NOT JUST AN IT ISSUE, IT'S A BUSINESS ISSUE, A HUMAN ISSUE, AND A **NATION'S ISSUE**.”

”

SECURE TODAY, PROTECT TOMORROW



REPORT IT

If you see something suspicious, report it to your organization.



CYBER SECURITY



QUIZ

Test your knowledge about Cyber Security!

1 What is Cyber Security?

- A. Protecting computers and networks from digital attacks
- B. Hacking into computer systems
- C. Creating viruses and malware
- D. Using the internet for shopping



2 Which of the following is an example of strong password?

- A. 12345678
- B. password
- C. P@ssw0rd!2024
- D. admin123



3 Phishing is a type of attack that tricks users into:

- A. Installing software updates
- B. Revealing sensitive information
- C. Backing up their data
- D. Increasing internet speed



4 Which of the following is NOT a type of malware?

- A. Virus
- B. Worm
- C. Trojan Horse
- D. Firewall



5 Which practice helps protect against ransomware attacks?

- A. Clicking on suspicious links
- B. Using strong antivirus and backups
- C. Sharing passwords with friends
- D. Disabling the firewall



6 What does 2FA stand for?

- A. Two Factor Authentication
- B. Two Firewall Access
- C. Two Factor Adjustment
- D. Two Step Authorization



7 Which of the following is a sign of a phishing email?

- A. It is from a trusted source
- B. It has urgent language and suspicious links
- C. It is well formatted
- D. It contains your correct personal details



8 Which port is commonly used for secure web browsing?

- A. 21
- B. 80
- C. 443
- D. 25



9 What should you do if you suspect your device is infected?

- A. Ignore it and continue using
- B. Disconnect from the internet and run a scan
- C. Format the device immediately
- D. Share the file with your friends



10 Which of the following is the best way to protect your data?

- A. Store it on a public computer
- B. Use weak passwords
- C. Encrypt your data
- D. Share it on social media



BE AWARE. BE SMART. STAY SECURE.

Cyber Security starts with you!



CRACK THE PASSWORD?



--	--	--

CODE
BREAKER

サイバー
ロック

A NUMERIC LOCK HAS A 3 DIGIT KEY

6 8 2

ONE NUMBER IS
CORRECT AND
WELL PLACED

6 4 5

ONE NUMBER IS
CORRECT BUT
WRONGLY PLACED

2 9 6

TWO NUMBERS ARE
CORRECT BUT
WRONGLY PLACED

7 3 1

NOTHING
IS CORRECT

8 2 4

TWO NUMBERS ARE
CORRECT BUT
WRONGLY PLACED



HINTS

- DIGITS ARE FROM 0 TO 9.
- EACH DIGIT IS USED ONLY ONCE.
- READ EACH CLUE CAREFULLY.

CAN YOU
CRACK IT?



LEARN TODAY
LEAD
TOMORROW

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PRACTICAL
PROJECTS



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